During the seminar the idea of developing a multi-criteria sustainable recommender system of Points of Interest to be visited in a city was discussed. The main point is that the recommender system not only takes into account the preferences of the user, but also other social, cultural, economic or environmental aspects, with the final aim of promoting responsible and sustainable tourism, reducing overcrowding and its associated negative effects, and improving the distribution of tourists and their wealth through the city.

Imagine that this recommender system has already been developed. You can take Barcelona as the city of reference, or any other city of your choice. You can consider at least 50 POIs distributed through the city. Your task in this seminar is to **design and develop a system for the evaluation of the effectiveness of this novel recommender system**, by comparing it to other basic options. The objective is to check, through the use of an **agent-based simulator,** whether the use of this system actually improves the management of tourists in the city (e.g. by reducing the usual overcrowding in the most popular spots and the accumulation of tourists in certain neighbourhoods). You can use any agent-based framework of your choice. The basic steps to be followed by the simulator could be the following:

- Generate a population of tourists (you could try to consider thousands of tourists, so that the final results are statistically significant).
- Assign each tourist a profile (that could include things like interests, budget, mobility mode, tolerance for walking, crowd aversion, sustainability sensitivity, preference for outdoor activities, travel with kids, etc.).
- Generate candidate POIs for each tourist. You should consider different recommenders (e.g. popularity-based recommender, interests-based personalized recommender, and the discussed multi-criteria sustainability-aware recommender).
- Simulate which POIs tourists actually visit.
- Update crowding levels of each POI and neighbourhood-level indicators.
- Compare outcomes across recommender strategies (you can even analyze the movement of the tourists from one POI to the next recommended one, if you want). In the discussion of the results you can consider different aspects commented on the seminar, like the precision/recall/diversity of the recommendations, or the fairness and trustworthiness of the recommender systems.

As an example, if we have a tourist interested in religious buildings, we could have the following:

- The popularity-based recommender could suggest Sagrada Familia, Casa Batlló, Parc Güell, Mercat de la Boqueria and Barri Gòtic, which are the top places in Tripadvisor. Sagrada Familia could be omitted if the budget is very low, for example.
- The interests-based recommender could suggest religious buildings like Sagrada Familia, Santa Maria del Mar, the Cathedral and the temple at the top of Tibidabo (Sagrado Corazón de Jesús). Sagrada Familia could be omitted if the user had crowd aversion, for example. The Tibidabo temple could be omitted if the user does not have tolerance for walking, for example.
- The sustainability-aware recommender could recommend the Sagrado Corazón de Jesús temple, the Ciutadella park, el Mercat dels Encants or the History Museum of Catalonia (combining the interest in religion buildings with aspects like the use of natural parks, the promotion of local markets and culture, and the distribution of tourists all around the city).

To deliver: report (in PDF format) and a link to the code/videos/files you have generated.